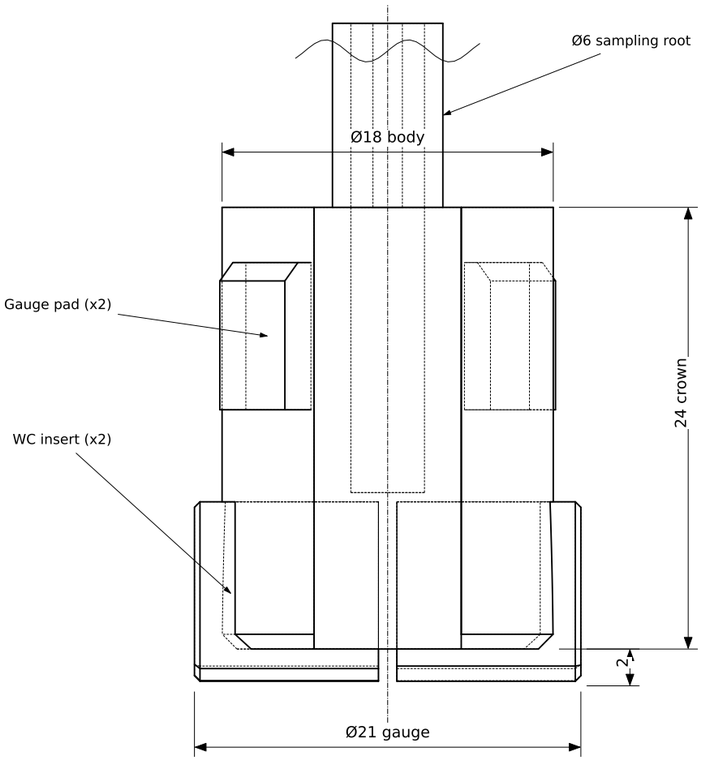
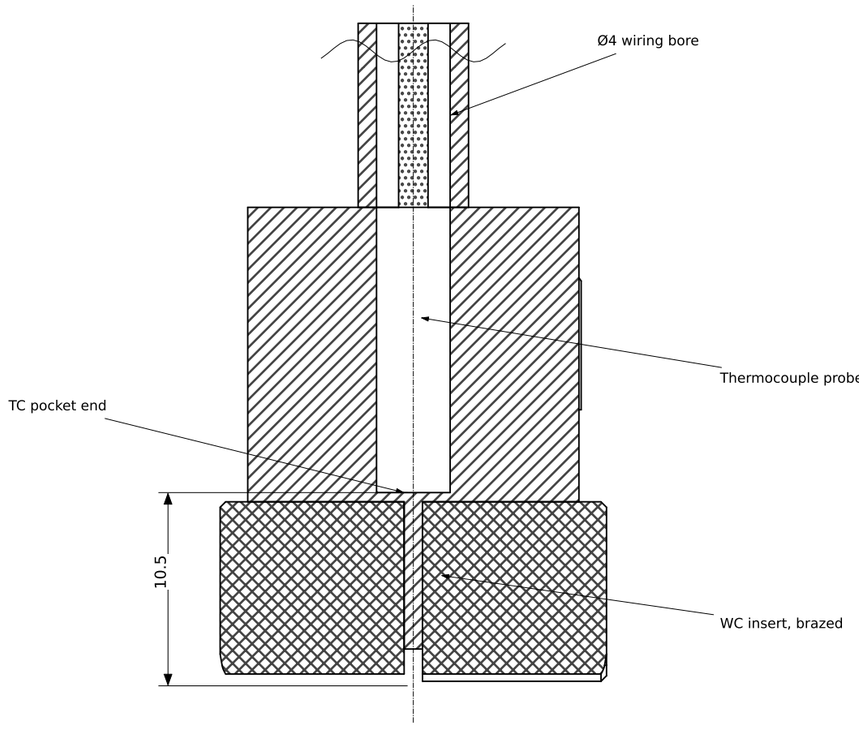


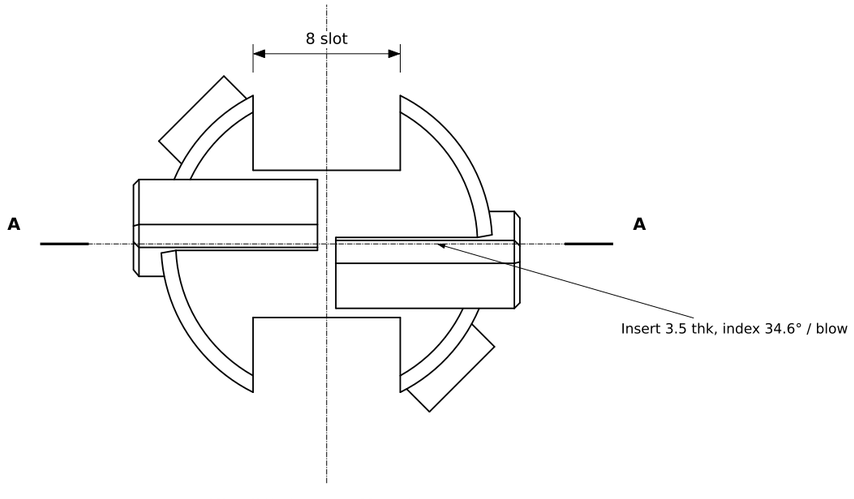
FRONT VIEW

BIT SUB ASSEMBLY 1:1



SECTION A-A





BOTTOM VIEW (cutting face)

NOTES

1. Inserts: WC-10Co cemented carbide (ISO K20), 3.5 x 10 mm chisel, silver-brazed, 50 um joint.
2. Rake -10 deg, clearance 15 deg, edge hone R1.0 (derived: Hertz contact on a basalt clast).
3. Gauge pads WC; crown body 1 mm under gauge; 8 mm junk slots ahead of each insert.
4. Joint M12x1 RH, 12 mm engagement; impact passes through the 16 mm shoulder.
5. Type-T thermocouple pocket ends 0.5 mm above the insert tops.
6. General tolerances ISO 2768-mK. Break sharp edges 0.2 unless shown.

DRILL BIT CROWN & BIT SUB

Autonomous low-gravity rotary-percussive bite-drilling system

Material: Body 17-4PH H1025 (same as auger)	Scale: 3:1 (bit sub 1:1) Units: mm Sheet: A3	
Drawn: Jay Tilak P. Date: 2022-09-07	DWG NO. **DRL-001**	REV A